Zane Collins

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EDUCATION

Harvey Mudd College, B.S. Physics (astrophysics focus) Aug 2023 – May 2027 (expected)
Coursework: Optics Laboratory, Electronics Lab, Engineering Systems, Observational Astrophysics, E&M Theory & Optics, Quantum Physics

RESEARCH EXPERIENCE

MIT, Seager Group — *Research Assistant, Solar Gravitational Lens Spectroscopy* May 2025 – Present

- Built a Python spectroscopic modeling pipeline largely from first principles, using HITRAN HAPI for high-resolution molecular line data and colorednoise to simulate noise in synthetic observations.
- Quantified biosignature detectability limits versus spatial resolution, spectral resolution, and signal-to-noise using cross-correlation and log-likelihood analysis.
- First-author poster at AASTCS 11: Exoplanet Atmospheres (2026); first-author manuscript in preparation.

MIT, Seager Group — *Research Assistant, Magma Oceans & Exoplanet Atmospheres* May 2026 – Present

- Coupling LavAtmos, FastChem, and petitRADTRANS to connect magma ocean interiors to observable atmospheric spectra; contributing author on a JWST Cycle 6 proposal for a young sub-Neptune.

MIT, Morning Star Missions to Venus — *Research Assistant* Jan 2025 – Aug 2026

- Assisted with a UV phase-modulation experiment, using a lock-in amplifier, to test in-situ photonic measurement of the Venusian atmosphere.
- Curated a spectral library of 10 molecule classes relevant to Venus; presented to collaborators at the University of Bern and mentored two high school students.

Harvey Mudd College — *Undergraduate Research Assistant, Applied Math* Aug 2024 – Dec 2024

- Developed quantile-based multiplicative update rules for corruption-robust non-negative matrix factorization and analyzed convergence under sparse noise; peer-reviewed paper at SampTA 2025.

LAB & INSTRUMENT PROJECTS

Table Mountain Observatory Sky Brightness — *Observational Astrophysics* Fall 2025

- Designed a month-long, three-field observing plan for the Pomona 1-meter telescope to characterize light pollution from Los Angeles and the Inland Empire as a baseline for new instruments.
- Built a Python reduction pipeline (calibration frames, plate solving, HST Guide Star Catalog zero points); validated results against the lunar cycle and cross-checked against an SQM sensor.

Spectrometer Alignment & Calibration — *Optics Laboratory* Feb – Apr 2026

- Aligned a Michelson Fourier transform spectrometer that resolved the sodium doublet; diagnosed a mid-experiment signal drop with photodiode tests, keeping compromised data out of the result.
- Calibrated a SPEX 500M grating spectrometer to 0.014% against 11 mercury lines and measured Rydberg constants within 0.04% (D) and 0.14% (H) of accepted values.

Standalone AM Radio — *Electronics Lab module, extended as a personal project* Fall 2026

- Replaced the lab-provided supply, antenna, and amplified speakers with my own: a power supply from a repurposed USB cable, an antenna hung from the highest dorm on campus, and an 8 Ω speaker amplifier; received a live AM station.

HST Globular Cluster Distance — *Observational Astrophysics* Fall 2025

- Built a ~150,000-source catalog from archival HST/ACS images, cut crowded sources and median-binned to reduce scatter, and fit isochrones to estimate a distance of 12.6 ± 0.6 kpc.

PUBLICATIONS & PRESENTATIONS

- Z. Collins, J. Haddock, T. Headley, L. Wang. "Quantile Multiplicative Updates for Corruption-Robust Nonnegative Matrix Factorization." SampTA 2025.
- **Z. Collins** et al. "Solar Gravitational Lens Telescope Extreme Spectroscopic Resolution." Poster, AASTCS 11: Exoplanet Atmospheres, BAAS 58(3), 2026.
- **Z. Collins**, A. Glidden, S. Barat, S. Seager, S. Turyshev. "Global High-SNR High-Resolution Atmospheric Spectroscopy Capability With the Solar Gravitational Lens Telescope." In preparation.

SKILLS & TEACHING

Programming: Python (NumPy, SciPy, Astropy, Matplotlib, Pandas), HITRAN HAPI, petitRADTRANS, FastChem, LavAtmos, Igor Pro, LabVIEW, Unix, LaTeX
Lab: lock-in amplifier, grating spectrometer, photomultiplier tube, Michelson interferometer, photodiode/amplifier, oscilloscope, function generator, spectrum analyzer, 1-m telescope
Methods: CCD reduction, aperture photometry, plate solving, photometric calibration, spectroscopy, cross-correlation, uncertainty analysis
Teaching: Physics & astrophysics lab TA and course grader, Harvey Mudd & Pomona Colleges, Jan 2025 – Present